Penalized Portfolio Optimization

Victor Ding

New Canaan High School

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Abstract

This research reviews practical considerations of long-short portfolio optimization problems. It studies a few penalty forms to address the limitations of pure constraints.

The Alpha Gate penalizes the excess turnover over a soft threshold before being strictly bounded by a higher limit. The regularization penalties we considered have parallels with regressions. Ridge regression [6] shrinks coefficients smoothly; Lasso [12] promotes sparsity through an L_1 penalty; and the Elastic Net combines the two. Unlike Elastic Net where both $L1$ and $L2$ penalties are simultaneously effective, the Berhu function [14] imposes $L1$ penalty on coefficients below a threshold, and $L2$ beyond that.

1 Introduction

Modern portfolio construction is fundamentally an optimization problem. Typically, a starting portfolio of existing positions is adjusted to a final portfolio. Position limits, attribute-exposure limits, and turnover budgets are standard constraints in quantitative portfolio management.

Each constraint imposed on a portfolio shrinks the feasible set of solutions, leading to a common concern that constraints "drive" the solution: an optimizer may satisfy them with little consideration of the objective function. A hard cap on turnover, for example, treats a small violation and a large one identically, potentially preventing a portfolio from considering strong signals whose payoff exceeds transaction costs. Many small trades are artifacts of estimation noise or are too small to justify transaction costs. These problems become even more prominent when accounting for estimation errors and data noises in the model inputs. Simply imposing hard constraints is overly simplistic and thus not very satisfactory.

This paper primarily studies penalty mechanisms for softening these rigidities. Our research proceeds as follows. We outline the framework in Section 2, where we review the various technicalities and practicalities of constrained portfolio optimization. We formalize the Alpha Gate as a soft turnover cap in Section 3. Section 4 covers Elastic Net and Berhu penalties used to prune noise trades. In Section 5, we use numerical experiments to illustrate the features and effectiveness of these penalty forms. Along the way, we also discuss the considerations for the choice of various penalty parameters. We highlight some important outstanding issues for future research in Section 6.

2 Implementation Framework

Modern portfolio construction involves optimizing an objective function defined over a vector of n asset positions $w \in \mathbb{R}^n$, where positive (negative) values represent long (short) positions. An example of a classic objective function is Markowitz's Mean-Variance framework:

$$\min_w \left[\frac{\lambda}{2} w' \Sigma w - r' w \right], \tag{1}$$

where r is a vector of expected returns, λ is an exogenous risk-aversion parameter, and Σ is a covariance matrix of asset returns. The design is driven by the desire to strike a balance between risks and payoffs.

Objective functions can take other forms, such as the squared L_2 from a target vector $\tilde{w} \in \mathbb{R}^n$. A given vector $\tilde{w}$ can be a benchmark portfolio to track, or a target portfolio driven by some forecasting models. It is often desirable to mimic such a portfolio as closely as possible. One reasonable approach is to minimize this objective:

$$\min_w \|w - \tilde{w}\|_2^2.$$

A positive semi-definite covariance matrix Γ can also be incorporated to generalize the above objective function further:

$$\min_w \left[\frac{1}{2}(w-\tilde{w})'\Gamma(w-\tilde{w})\right] = \min_w \left[\frac{1}{2}w'\Gamma w - \tilde{w}'\Gamma w\right]. \tag{2}$$

Mathematically, this expression is a special case of (1).

A majority of portfolio construction problems can be formulated as optimizing certain quadratic forms like the above subject to some linear constraints. While potential payoffs are often uncertain, transaction costs are incurred upon execution. Turnover budgets and individual turnovers are often imposed, so a solution is discouraged from chasing marginal benefits with excessive transaction costs. Other typical constraints include limit on risk exposures.

Objective function (1) is convex if Σ is positive semi-definite. The optimization problem of such an objective function has a global optimal solution if the constraints also maintain a convex feasible set.

Let us focus on minimizing the family of objectives with a quadratic term $q(w)$, which captures the conventional Mean-Variance problem (1) and target tracking problem (2). Such objectives are within the reach of Disciplined Convex Programming (DCP). DCP problems can be easily solved with DCP solvers.

Linear equalities, and inequalities define hyperplanes and half-spaces respectively. Half-spaces, along with second-order cones, define convex sets. Hyperplanes are affine, and thus convex sets. When feasible sets are convex, optimization is often far more efficient and accurate. Because of this, we show a preference for preserving convexity of the overall feasible region. For constraints that are non-smooth but inherently convex (such as absolute value or piecewise functions), we use exact epigraph reformulations with auxiliary variables. For constraints that are truly non-convex, we employ convex restrictions to approximate the feasible space while maintaining computational tractability.

Inputs such as return estimates, risk model coefficients, and target forecasts in $\tilde{w}$ are all estimates subject to estimation errors and data noise. Thus, a problem subject to overly permissive constraints is susceptible to concentrated positions with potentially poor signals. On the other hand, we also look to avoid overly tight constraints. While constraints have their purposes and those like turnover and position limits may even serve as a natural form of regularization on model solutions, each one shrinks the feasible set of potential solutions. Tight constraints may force solutions that reflect little of their intended objective functions. Ideally, constraints should be set such that they do not "drive" the solution; instead a solution should be dictated primarily by the objective function.

Hard constraints that bound turnover and other portfolio characteristics have certain downsides. We prefer to impose them to reflect practical and risk limits, rather than determining optimality. Violating a hard constraint by a negligible amount and by a comparatively huge amount has the same effect, meaning rounding errors, tiny violations, and genuine infeasibility are treated equally.

Furthermore, some powerful signals may warrant trading significantly more, either of individual assets or across the entire portfolio when the risk-adjusted payoffs far exceed the likely incurred transaction costs. In those cases, it is desirable to permit certain violations of hard constraints. To achieve this, we introduce soft constraints.

Soft constraints allow breaches of some thresholds while discouraging large ones by adding a penalty to the objective function. We will consider three distinct types of soft constraints: a turnover penalty, an Elastic Net, and a Berhu penalty.

2.1 Constraints

Position limits can be represented as

$$\underline{w} \leq w \leq \overline{w},$$

where constant vectors $\underline{w} \in \mathbb{R}^n$ and $\overline{w} \in \mathbb{R}^n$ are lower and upper bounds, respectively.

Similarly, limits on each individual turnover are represented by $|w - w^*| \leq t$, where constant $t \in \mathbb{R}^n$ defines individual turnover bounds, and w^* represents the starting weight vector. Turnover bounds can be essentially incorporated into position limits because w^* is a constant vector.

We define an attribute as a characteristic of an asset. It is common to cap portfolio exposures to sectors and industries. Equity style factors (growth, momentum, value, etc.) are also types of attributes widely used in risk modeling. An asset can have exposures to multiple attributes. For example, although a small tech startup can have exposure to just the technology sector, a conglomerate operates in several. Attribute exposures can also be negative. Smaller companies, for example, tend to have negative exposures to size factors.

We express n assets' exposure in m attributes with an $m \times n$ matrix $\mathbf{S}$. A hypothetical $\mathbf{S}$ for n assets across various attributes is shown in Table 1. The total portfolio exposures to the m factors are $\mathbf{S}w$. The exposure limits can be represented as

$$\underline{s} \leq \mathbf{S}w \leq \overline{s}, \tag{3}$$

where $\underline{s}$ and $\overline{s}$ are lower and upper exposure bounds, respectively.

Net exposure, equal to the value of long positions minus the value of short positions, is a special form of attributes. It can be captured as a row of ones in $\mathbf{S}$.

A constant upper bound $T > 0$ may be imposed on total turnover as

$$\|w^* - w\|_1 \leq T. \tag{4}$$

This introduces a total budget over the sum of individual turnovers. In practice, individual turnover limits t are often set such that $T < \sum_i t_i$; otherwise, the total turnover cap will become redundant.

Table 1: Example Factor Loading Matrix.

	Assets				
	1	**2**	**3**	...	n
Finance	0.10	0.40	0.20	...	A_n
Energy	0.13	0.20	0.30	...	B_n
Communication	0.06	0.04	0.10	...	C_n
...	...	...	...	...	...
Growth	0.70	0.30	0.20	...	D_n
Momentum	-1.0	-0.30	0.10	...	E_n
...	...	...	...	...	...

Total turnover can be further generalized with a weighting vector $c > 0$ such that

$$c'|w^* - w| \leq T. \tag{5}$$

The vector c and T may represent some transaction cost estimates and a budget limit, respectively. When turnovers are equally weighted, constraint (5) becomes (4).

The constraint (4) or (5) is not directly supported by many solvers. Although convex, it is not affine, and is not differentiable at w^*. We introduce the slack vector $f \in \mathbb{R}^n$, such that

$$|w^* - w| \leq f, \quad c'f \leq T. \tag{6}$$

A feasible f exists if and only if $c'|w^* - w| \leq T$. Hence, (6) and (5) define the same feasible set of w.

Gross exposure, $|w\|_1$, is another measure often involved in some constraints, as it is often desirable to keep a specific amount of capital invested in a portfolio. It is common to limit the gross exposure (leverage) by an upper bound G as

$$|w\|_1 \leq G. \tag{7}$$

This is still a convex constraint like that on total turnovers. But the optimal solution may find it cheaper to settle at a gross exposure far less than G, effectively underinvesting with available capital. It is thus attractive to place a lower bound $\underline{G}$ on the gross exposure as well. When $G = \underline{G}$, the constraint becomes an equality

$$\|w\|_1 = G. \tag{8}$$

In long-only or short-only portfolios, absolute value signs in (8) can be trivially removed so that it becomes a convex constraint. But the presence of a lower bound or equality constraints in long-short portfolios makes the feasible set non-convex, which poses a problem for most solvers. We will address this non-convexity in section 2.2.

2.2 Global Optimality Conditions and Difficulties of Gross Exposure

Because the portfolio construction problems considered henceforth are convex, their solutions are exact and can be calculated via DCP. Suppose a quadratic objective is bounded by affine constraints that create a convex feasible set. The Karush-Kuhn-Tucker (KKT) conditions [1] define the framework for determining optimality.

In an optimization setting with quadratic objective function, a positive semi-definite Hessian matrix ensures convexity of the contour set. When constraints are affine, the feasible set is also convex. Under such conditions, a local optimal minimum is also the global minimum.

A well-behaved covariance matrix Σ in (1) is positive semi-definite, so is Γ in (2). Almost all the constraints discussed so far are also convex except for gross exposure.

A set $\mathcal{C} \subset \mathbb{R}^n$ is convex if and only if for any two points a and b in $\mathcal{C}$ and every $\theta \in [0, 1]$ the following is true:

$$\theta a + (1 - \theta)b \in \mathcal{C}. \tag{9}$$

We will prove that for a fixed $G > 0$, the set $\mathcal{S} = \{w \in \mathbb{R}^n : \|w\|_1 = G\}$ is not convex. This argument does not restrict itself to properties of the L_1 norm. Rather, it uses two axioms every norm satisfies:

1. Absolute Homogeneity: $\|\alpha w\| = |\alpha|\|w\|$ for any scalar α.
2. Positive definiteness: $\|w\| = 0 \Leftrightarrow w = 0$.

Take any $w \in \mathcal{S}$. By homogeneity with $\alpha = -1$, we have

$$\|-w\|_1 = |-1| \cdot \|w\|_1 = \|w\|_1 = G.$$

So,

$$-w \in \mathcal{S}.$$

For $\theta = \frac{1}{2}$, we have $\theta w + (1 - \theta)(-w) = 0 \notin \mathcal{S}$. Therefore, the set $\mathcal{S}$ is not convex.

The non-convexity arises from the uncertain signs of the optimal solution, which are not known prior to the solution process. We can rewrite w as $w^+ - w^-$, where both w^+ and w^- are nonnegative vectors in $\mathbb{R}^n$. This allows us to represent G as $e'(w^+ + w^-)$. The problem is that $w_i^+ w_i^-$ must equal to 0 for all i or a solver will freely choose them to hit the target G. Even Mixed-Integer Convex Programming (MICP) will have difficulty finding an optimal solution in a timely and resource-efficient manner when n is reasonably large.

One way to overcome this is forcing the signs of $\{w_i\}$ according to some rules prior to setting up the gross exposure constraint. Let $G^+ > 0$ be the desired long exposure, and $G^- < 0$ be the negative counterpart. The total gross exposure becomes $G^+ - G^- = G$. For tracking problems of

(2), we propose to match the sign of w to that of $\tilde{w}$ for all i such that

$$\begin{aligned}
\sum_{i\in\mathcal{W}^+} w_i = G^+, \qquad & \mathcal{W}^+ = \{i|\tilde{w}_i > 0\}, \\
\sum_{i\in\mathcal{W}^-} w_i = G^-, \qquad & \mathcal{W}^- = \{i|\tilde{w}_i < 0\}, \\
\sum_{i\in\mathcal{W}^0} w_i = 0, \qquad & \mathcal{W}^0 = \{i|\tilde{w}_i = 0\}. \\
w_i \geq 0, \qquad & i \in \mathcal{W}^+. \\
w_i \leq 0, \qquad & i \in \mathcal{W}^-. \\
w_i = 0, \qquad & i \in \mathcal{W}^0.
\end{aligned} \tag{10}$$

The price is that the feasible set is effectively shrunk, and consequently may not yield the same global optimal solution of the original problem.

There are other ways to eliminate non-convexity. Other local search methods may improve upon this approach:

- Solve the original problem with inequality (7), and use the signs of the solution vector to derive $\mathcal{W}^+$, $\mathcal{W}^-$, and $\mathcal{W}^0$. Solve the problem with the equality constraint along with the determined signs.
- After one iteration, flip the signs of assets with $|w_i|$ inside a small band ϵ to update $\mathcal{W}^+$, $\mathcal{W}^-$, and $\mathcal{W}^0$. Solve again. If there is improvement, increase ϵ and repeat. If not, stop.
- Solve the original problem with inequality (7), and scale all the weights such that $\|w\|_1 = G$. Feed the results back to iterate.
- An approach in the same spirit as the penalty forms discussed later is to introduce a slack variable $g \geq 0$ such that $||w||_1 + g = G$, and a penalty term ηg. A sufficiently large parameter η will cause g to be 0 at the optimal solution. Fix the signs according to the solutions of this penalized optimization and solve the original problem without the penalty term.

Unfortunately, these are all heuristic approaches to resolve sign uncertainty prior to the optimization. None of these approaches can guarantee the global optimum with these approaches. As long as the feasible set is not expanded like (7), the objective functions across various heuristic approaches can be compared to each other to evaluate the validity of each.

3 Hard and Soft Caps of Transaction Costs

Trading from w^* to w incurs transaction costs. It is thus desirable to balance the objective function and turnover to avoid paying deterministic upfront costs for insignificant or uncertain transactions. Imposing a hard budget limit on total turnover, however, handicaps portfolio construction from achieving its objective. Instead, we should explore the potential trade-offs of excessive turnovers.

In addition to a hard upper bound T on total turnover (4) or as its replacement, we introduce a slack variable $z >= 0$ to relax a constant soft bound $\tilde{T} > 0$ such that

$$\|w - w^*\|_1 \leq \tilde{T} + z,$$

and add a penalty term to the objective function $q(\cdot)$,

$$p(z) = \alpha z^2. \tag{11}$$

This term will be referred to as Alpha Gate. The exogenous parameter $\alpha > 0$ determines the cost of going above $\tilde{T}$. The variable z will be optimized along with w. The quadratic term has the benefit of having a comparable magnitude with many objective function forms of $q(w)$ such that the choice of α is more intuitive. It is completely reasonable to replace it with $\alpha|z|$. The presence of an absolute value poses no technical difficulty here given that $z \geq 0$.

This formulation is very intuitive. The total turnover is allowed to exceed $\tilde{T}$, but pays a penalty to do so. If turnover is less than $\tilde{T}$, no penalty is paid. The larger the value of α, the more strongly the total turnover is discouraged from exceeding $\tilde{T}$, and vice versa. This "soft cap" is not a rigid hard bound, allowing some room to trade more when situation dictates it. When both a "hard cap" and a "soft cap" are present, we generally have $\tilde{T} < T$ in that T serves as a rigid safeguard on the total turnover.

4 Regularization of Noisy Trades

While individual turnover and transaction budget constraints are common in portfolio construction, it is also desirable to reduce the number of inconsequential small trades. They are either results of some computational or statistical noises, or are simply too insignificant to justify the execution or handling costs. Eliminating such small trades also gives the turnover budget to other assets.

We wish to induce a degree of sparsity in a portfolio's turnovers, pruning noisy trades. A popular method to encourage sparsity involves setting a certain threshold for $|w_i - w_i^*|$, below which those $\{w_i\}$ will be forced to stay at $\{w_i^*\}$, meaning no trade allowed. However, this ad-hoc rule may be too rigid, and the threshold setting also seems rather arbitrary.

This problem is very similar to variable selection in various regression frameworks where coefficients of some variables are shrunk towards zero to yield parsimonious models. It is thus not surprising that similar techniques can be employed to regularize $|w - w^*|$.

4.1 Elastic Net

Ridge Regression [6] softens estimated coefficients through a scaled L_2 norm. Lasso regression [12] utilizes an L_1 norm instead. The Elastic Net [14] uses both L_1 and L_2 norms simultaneously. This

concept can be adopted for portfolio optimization, penalizing turnovers:

$$\gamma \left[\beta \|w - w^*\|_1 + \frac{1-\beta}{2} \|w - w^*\|_2^2 \right], \tag{12}$$

where $0 \leq \beta \leq 1$ and $\gamma > 0$ are adjustable exogenous parameters. It can be added to a base quadratic objective $q(\cdot)$ along with other penalty terms such as (11). Each term in the objective is either an L_1 norm, L_2 norm, or a quadratic function. Since all scalars α, β, and γ are positive, the entire objective function maintains its convexity.

The L_1 penalty promotes sparsity, driving small turnovers to 0. The L_2 penalty promotes smoothness. It prevents related trades from swinging violently with tiny input changes. Turnover among correlated assets is distributed more evenly than with only an L_1 penalty. For example, the L_2 penalty resulting from two turnovers 0.2 and 0.2 is smaller than the penalty resulting from 0.1 and 0.3, even though the total turnover is the same. The parameter β governs relative strength of penalties between the two terms.

The parameter γ controls the overall scale of this penalty term relative to the objective function $q(\cdot)$. A larger $\gamma\beta$ reflects a higher marginal cost of initiating a new trade, pushing some small trades towards zero.

4.2 Berhu (Reverse Huber) Loss

Elastic Net penalties in the last section subject trades to both L_1 and L_2 penalties, regardless of the magnitude of a trade. It tends to reduce turnovers indiscriminately. We introduce the Berhu (Reverse Huber) loss, a piecewise function that contains either an L_1 or L_2 penalty over non-overlapping regions.

Denote $q_i = |w_i - w_i^*|$. Frequently used in regressions, the Huber's loss function takes the form

$$h(q_i) = \begin{cases} \frac{q_i^2}{2\epsilon} & \text{if } q_i \leq \epsilon, \\ q_i - \frac{\epsilon}{2} & \text{if } q_i > \epsilon. \end{cases} \tag{13}$$

It applies a quadratic penalty for small deviations, and a linear penalty for large deviations. This dampens the influence of extreme outliers in regression while maintaining differentiability throughout.

Berhu Loss, also known as Reverse Huber's loss, inverts the order of the quadratic and linear functions. Given $y_i = w_i - w_i^*$ and a threshold scalar $\epsilon > 0$, the Berhu Loss is defined as

$$b(y_i) = \begin{cases} |y_i| & \text{if } |y_i| \leq \epsilon, \\ \frac{y_i^2 + \epsilon^2}{2\epsilon} & \text{if } |y_i| > \epsilon. \end{cases} \tag{14}$$

The Berhu function works similarly to Elastic Net, but only applies either an L_1 norm or L_2 norm at a time. For minor weight adjustments where $|y_i| \leq \epsilon$, the penalty is linear, promoting

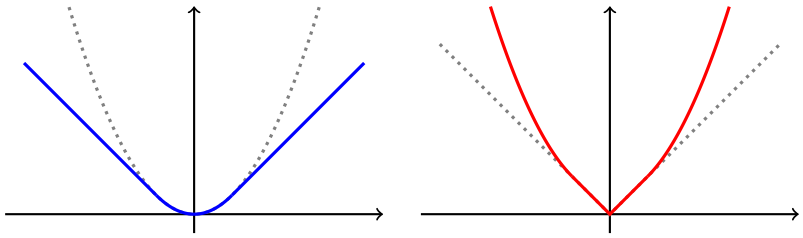


Figure 1: Huber's Loss (Left) and Berhu Loss (Right)

sparsity. For significant weight adjustments where $|y_i| \geq \epsilon$, the penalty becomes quadratic and tend to spread trades across assets.

Intuitively, Elastic Net sharpens but does not completely eliminate the smooth edges of the contour set while the Berhu function shrinks the areas of pointy edges but still preserves the non-differentiable vertices. Both encourage more control variables to go towards zero upon contact with the objective contour set at the optimal solution, just in slightly different fashions. Once a turnover exceeds ϵ, the Berhu Loss becomes quadratic while the Elastic Net still maintains an L_1 term. Therefore, the Berhu Loss penalize the large outliers more aggressively than the similar Elastic Net functions, while Elastic Net steadily pushes all the trades toward zero via its $L1$ term. A properly chosen ϵ can potentially deliver $L1$ penalties on some trades while exempting others.

One advantage that Elastic Nets maintains over Berhu functions is in its handling of correlated trades. Since only $L1$ penalty is effective for the trade magnitudes below ϵ, the Berhu shrinkage is highly sensitive to correlated trades that are also below ϵ.

The Berhu function is convex and continuous everywhere. That said, its piecewise nature makes it difficult to implement directly into a solver, as many solvers do not accept piecewise functions natively or have a designated Berhu function. To resolve this, we will create an epigraph formulation with auxiliary variables.

Define two non-negative vectors $u \geq 0$ and $v \geq 0$ subject to the following constraints for all i:

$$u_i + v_i \geq y_i, \quad u_i \leq \epsilon,$$

where y is defined in (6). The Berhu Loss then becomes

$$b(y) = \sum_i \left(u_i + v_i + \frac{v_i^2}{2\epsilon} \right). \tag{15}$$

When $y_i \leq \epsilon$, to minimize the penalty, the solver will set $v_i = 0$, since the quadratic penalty term $\frac{v_i^2}{2\epsilon}$ grows more rapidly than the linear term u_i, being more "expensive" in the objective function. The constraint $u_i + v_i \geq y_i$ forces $u_i = y_i$, making the penalty exactly y_i, matching the linear branch of (14). When $y_i > \epsilon$, the minimizer pushes u_i to its upper bound ϵ, resulting in v_i being non-zero $v_i = y_i - \epsilon$, while the cost becomes $\frac{y_i^2+\epsilon^2}{2\epsilon}$. This matches the value of the quadratic branch of (14).

A penalty term based on (15) can be defined as

$$b_t(y) = \gamma \sum \left(u_i + v_i + \frac{v_i^2}{2\epsilon} \right) = \gamma \sum_i \left[y_i + \frac{1}{2\epsilon} \max[y_i - \epsilon, 0]^2 \right], \tag{16}$$

where $\gamma > 0$ is an exogenous scale coefficient that determines the scale of this term relative to the objective function $q(\cdot)$ and other penalty terms present. All terms in the objective function introduced so far are either quadratic or linear, so the objective function remains convex.

Threshold ϵ is the knot where the penalty transitions from linear to quadratic. A small ϵ means that the optimizer switches to quadratic penalties at very small turnover values, likely resulting in fewer turnovers reduced to exactly 0. A larger ϵ means the penalty remains linear for a wider range of turnovers, penalizing those trades less aggressively. The choice of ϵ is highly sensitive to the data scales, and depends on the nature of the data.

5 Numerical Experiments

An ideal set of penalty parameters should exert the least amount of influence while achieving the intended objectives. For example, a set of regularization parameters for Elastic Net or Berhu penalty is ideal when they are the least aggressive while shrinking just enough number of trades to zero consistently. An Alpha Gate is ideal when it pushes the realized total turnover to narrowly fluctuate around a preferred stable level between $\tilde{T}$ and T.

If the target portfolio weights are derived with some stable modeling process, a set of penalty parameters can be evaluated based on its consistency over a sample period. In this section, we set out to study the behaviors of three penalty forms introduced so far.

We base our numerical experiments on a proprietary dataset provided by a hedge fund manager. The dataset contains 10 years of daily trading signals for the most liquid US stocks and their attribute matrices. There are three signal series intended as target portfolio weights, varying in natural turnovers. They were generated from some quantitative models that were not concerned with various portfolio construction considerations. Our objective is to implement long-short portfolios that replicate these target weights as closely as possible subject to various constraints. We focus on a special form of loss function (2) with Γ being an identity matrix,

$$\min_w \|w - \tilde{w}\|_2^2,$$

where $\tilde{w}$ represents a target model weight vector.

Let $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}_3$ represent the three signal series from this dataset. Table 2 contains their summary statistics. All the portfolio weights are net of their means and then normalized by the gross exposures. Because $\mathcal{D}_1$ has the highest natural turnover, the turnover constraints are most likely to bind for this series; we therefore use $\mathcal{D}_1$ as the main diagnostic series in the parameter studies below.

Table 2: Summary Statistics of the Raw Dataset (01/04/2016 – 12/31/2025)

	$\mathcal{D}_1$	$\mathcal{D}_2$	$\mathcal{D}_3$
Daily Average of Number of Qualified Stocks	2,201	2,201	2,201
Daily Average of Meaningful Positions ($> 10^{-8}$)	$1,223$	$1,667$	$1,862$
Daily Average of Total Turnover (%)	131.7	91.83	76.75
Daily Average of Position Standard Deviations	0.154	0.159	0.165
Daily Average of Max Position Sizes (%)	1.400	1.558	1.810
Daily Average of Max Change Magnitudes (%)	1.117	0.957	0.938

5.1 Experimental Design and Metric Definitions

To maintain comparability, we define a common set of constraints for all the experiments:

$$\|w|_1 = 2, \tag{17a}$$

$$e'w = 0, \tag{17b}$$

$$|w| \leq 0.04, \tag{17c}$$

$$|w - w^*| \leq 0.02, \tag{17d}$$

$$\|w - w^*\|_1 \leq 0.7, \tag{17e}$$

$$|\mathbf{S}w| \leq 0.001. \tag{17f}$$

The rules outlined in (10) are applied to eliminate the non-convexity due to the gross exposure constraint (17a). The net exposure constraint (17b), along with (17a), implies that both the total long and short book size are both equal to 1. Often portfolio weights are discussed in terms of long book sizes. But in our setup, all the portfolio units are normalized to 2 effectively. The position constraint (17c) limits all the positions below 4% of the long book size, while (17d) limits the individual trades below 2%. The total turnover constraint (17e) caps that at 70%. The attribute matrix $\mathbf{S}$ in (17f) has 30 rows, corresponding to the most important 30 risk attributes, which caps those exposures at 0.001. These constraints are quite common in actual trading environments.

For each experiment with a set of parameters and a signal series, we computed a few metric series on each day k:

- $o_k^{(0,i)}$: realized total turnover $\|w - w^*\|_1$;
- $l_k^{(0,i)}$: realized loss function $\|w - \tilde{w}\|_2^2$;
- $c_k^{(0,i)}$: correlation between the optimal solution w and target $\tilde{w}$;
- $z_k^{(0,i)}$: portion of optimized trades with magnitude below 10^{-8}.

We ran a baseline iteration with these standard constraints for signal series $\mathcal{D}_i$ where $i = 1, 2, 3$. The results are presented in table 3. All three signals have natural turnovers above the limit of 70%, causing the constraint (17e) to be binding across the board. The constraints also push the

Table 3: Baseline Results

	$\mathcal{D}_1$	$\mathcal{D}_2$	$\mathcal{D}_3$
Average absolute position size (%)	0.035	0.035	0.035
Average $\{o_k^{(0,i)}\}$	0.700	0.700	0.700
Average $\{c_k^{(0,i)}\}$	0.961	0.984	0.988
Average $\{z_k^{(0,i)}\}$	0.452	0.250	0.162
Average $\{l_k^{(0,i)}\}(\times 10^{-4})$	4.068	1.687	1.319

correlation between the solution weights and the target weights away from 1. Since the natural signal turnovers decrease from $\mathcal{D}_1$ to $\mathcal{D}_3$, the hard turnover limit places decreasing pressure on the solution. This is evident from the correlation between the solution weights and the target weights, and the realized loss functions. There is decreasing distortion from the target in the solutions from $\mathcal{D}_1$ to $\mathcal{D}_3$.

For the experiments involving the Elastic Net penalty (12), let $e(\beta, \gamma)$ be the experiment with parameter (β, γ). We represent the corresponding metric series as $\{l_k^{(e,i)}(\beta, \gamma)\}$ (without the penalty term), $\{c_k^{(e,i)}(\beta, \gamma)\}$, and $\{z_k^{(e,i)}(\beta, \gamma)\}$, respectively. Similarly for Berhu penalty (16), we have $\{l_k^{(b,i)}(\epsilon, \gamma)\}$, $\{c_k^{(b,i)}(\epsilon, \gamma)\}$, and $\{z_k^{(b,i)}(\epsilon, \gamma)\}$, respectively. For those involving the alpha penalty (11), we have $\{l_k^{(a,i)}(\alpha)\}$, $\{c_k^{(a,i)}(\alpha)\}$, and $\{o_k^{(a,i)}(\alpha)\}$ instead of $\{z_k^{(a,i)}(\alpha)\}$.

Based on these metric series, we calculated the following averages over d days:

$$L^{(e,i)}(\beta,\gamma) = \frac{1}{d}\sum_{k=1}^{d}\Big[l_k^{(e,i)}(\beta,\gamma) \div l_k^{(0,i)}\Big],$$

$$C^{(e,i)}(\beta,\gamma) = \frac{1}{d}\sum_{k=1}^{d}\Big[c_k^{(e,i)}(\beta,\gamma) - c_k^{(0,i)}\Big],$$

$$Z^{(e,i)}(\beta,\gamma) = \frac{1}{d}\sum_{k=1}^{d}\Big[z_k^{(e,i)}(\beta,\gamma) - z_k^{(0,i)}\Big],$$

$$L^{(b,i)}(\epsilon,\gamma) = \frac{1}{d}\sum_{k=1}^{d}\Big[l_k^{(b,i)}(\epsilon,\gamma) \div l_k^{(0,i)}\Big],$$

$$C^{(b,i)}(\epsilon,\gamma) = \frac{1}{d}\sum_{k=1}^{n}\Big[c_k^{(b,i)}(\epsilon,\gamma) - c_k^{(0,i)}\Big],$$

$$Z^{(b,i)}(\epsilon,\gamma) = \frac{1}{d}\sum_{k=1}^{n}\Big[z_k^{(b,i)}(\epsilon,\gamma) - z_k^{(0,i)})\Big],$$

$$L^{(a,i)}(\alpha) = \frac{1}{d}\sum_{k=1}^{d}\Big[l_k^{(a,i)}(\alpha) \div l_k^{(0,i)}\Big],$$

$$C^{(a,i)}(\alpha) = \frac{1}{d}\sum_{k=1}^{d}\Big[c_k^{(a,i)}(\alpha) - c_k^{(0,i)}\Big],$$

$$O^{(a,i)}(\alpha) = \frac{1}{d}\sum_{k=1}^{d}\Big[o_k^{(a,i)}(\alpha) - o_k^{(0,i)}\Big].$$

The loss ratio $L^{(\cdot,i)}$ measures the realized impact on the objective function of a penalty term for $\mathcal{D}_i$. A bigger value indicates that the penalty pushes the solutions further away from the original objective. The correlation difference $C^{(\cdot,i)}$ is another measure of the impact of a penalty function. An ideal portfolio should have a high correlation with the original signal, while a penalty term will likely decrease such a correlation. The difference is likely to be negative and decreases further as the penalty becomes more aggressive. The zero trade difference $Z^{(\cdot,i)}$ measures the effectiveness of a penalty term in trimming noise trades. The turnover difference $O^{(\cdot,i)}$ measures the effectiveness of a penalty in damping the total turnover below the hard threshold. These metrics may also serve as the criteria in choosing penalty parameters, where they are evaluated in a sample period.

For each parameter setting, we ran optimizations over the entire sample period for all the three signal series. They exhibit similar patterns. We shall base our discussions on the experiments on $\mathcal{D}_1$. The corresponding results of $\mathcal{D}_2$ and $\mathcal{D}_3$ are shown in the appendix.

In setting parameters for experiments, we start with a coarse set of parameters over a wide range. After each batch run, we then focus on the range of parameters that have more desirable properties with finer search grid of parameters. A few iterations are usually carried out. This helps reduce the search space, but also results in uneven concentration of data points over some areas of

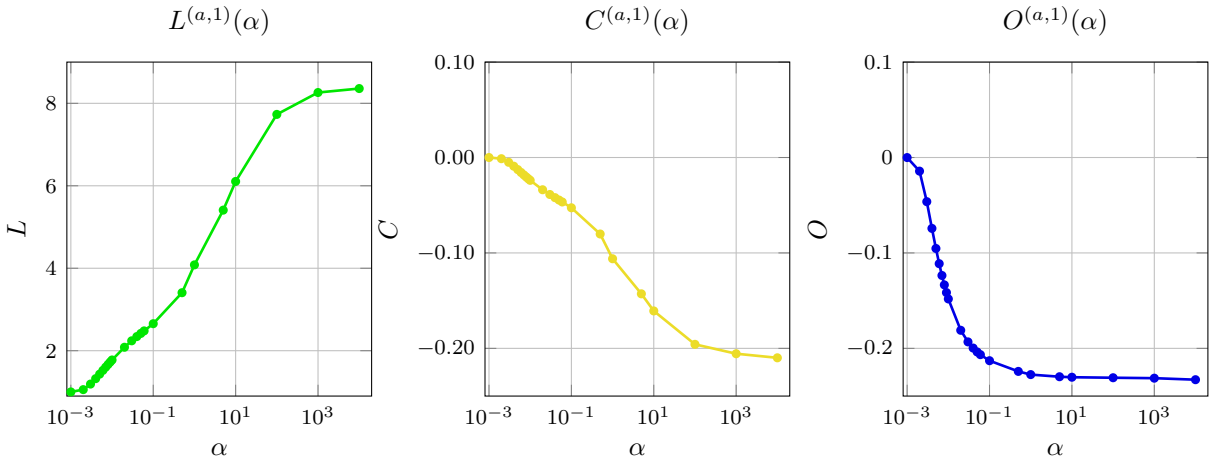


Figure 2: Alpha Gating $\mathcal{D}_1$

the parameter spaces.

5.2 Impacts of Alpha Gating

Given the hard total turnover limit 0.7, we fix the soft bound $\tilde{T}$ at 0.4 for our experiments of alpha gating. The α parameter in (11) dictates the relative importance between the loss function and the quadratic excess of total turnover over the soft threshold. The loss function is basically a sum of squared individual turnovers from the target weights, while the penalty term is a squared total turnover from the initial positions in excess of a constant. Since $\|w - \tilde{w}\|_2^2 < \|w - w^*\|_1^2$, a unit of α is more significant than a unit of change in the loss function.

Plots of metrics $L^{(a,1)}(\alpha)$, $C^{(a,1)}(\alpha)$, and $O^{(a,1)}(\alpha)$ in relation to the penalty parameter α are shown in Figure 2.

The penalty starts kicking in once the total turnover exceeds 0.4. All three metrics behave as expected. Both $C^{(a,1)}$ and $L^{(a,1)}$ keep deteriorating until the total turnover approaches the soft bound at 0.4 when the penalty term approaches zero. The loss ratio $L^{(a,1)}$ rises sharply at first and then decelerates where $O^{(a,1)}$ plateaus. Initially, $O^{(a,1)}(\alpha)$ decreases precipitously as α increases, then later asymptotes at a value of around -0.24. This implies the realized turnover approaches 0.46. Considering that $T = 0.7$ and $\tilde{T} = 0.4$ and that constraints are binding without the penalty, Alpha Gating successfully acts as a soft barrier. The values of α used in the experiments are largely concentrated between 10^{-3} and 10^{-1} to illustrate more rapid change over that range.

The asymptotes in the three graphs reveal that increasing α beyond ≈ 0.1 results in close to no reduction in turnover $O^{(a,1)}(\alpha)$, but continues to slowly diminish correlation and increase loss. Past this point, further distorting the solution to cater to the aggressive penalty yields only small reduction in turnover. There will no longer be significant value in increasing α further if the goal is controlling turnover.

A drop in realized total turnover as much as 20% can be achieved with a relatively small α, corresponding to about 2% drop in $C^{(a,1)}$. At the same time, the realized loss function almost

doubles with such a penalty. The increase in the loss function is somewhat expected because reduction in turnovers will almost always increase the loss function whether the limit is soft or hard.

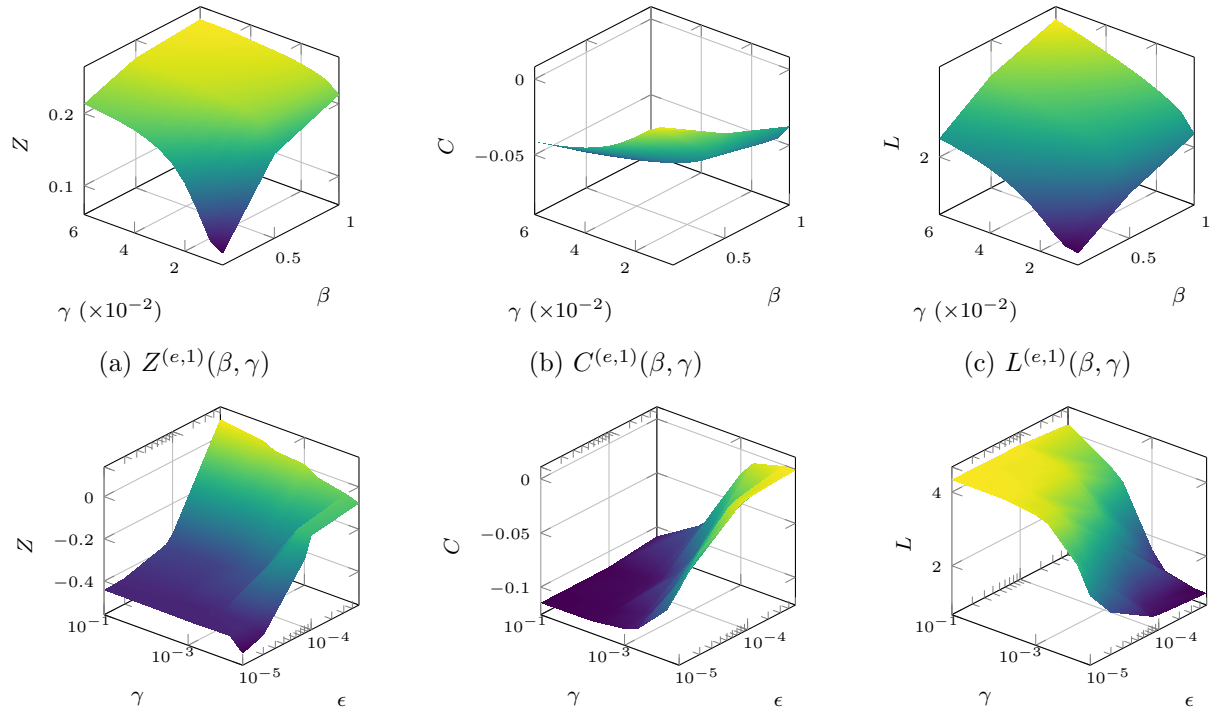
5.3 Impacts of Berhu and Elastic Net Penalties

Elastic Net is similar to dialing two switches. The L_1 penalty discourages small trades and the L_2 penalty, which is always active, spreads trades and discourages very large ones. The penalty applies both effects at each trade size. Meanwhile, Berhu is a pure L_1 penalty for small trades and a pure L_2 penalty for large ones.

For elastic graphs in Figure 3, a β range of $(0, 1]$ was selected, while γ was selected over a range of $[5 \cdot 10^{-3}, 6 \cdot 10^{-2}]$. A β of zero is not meaningful in pruning noise trades. A β of 1 is also unlikely to be desirable because a pure Lasso form is very sensitive to highly correlated features. In our use cases, some trades are bound to be correlated so that the solution may result in great variation in pruning zero trades.

For Berhu graphs, ϵ parameters are spread over $(10^{-5}, 5 \cdot 10^{-4})$, matching the typical target weights, while γ was selected within a range of $[5 \cdot 10^{-5}, 10^{-1}]$.

Graphs for various data points in both Berhu and Elastic Net iterations are displayed in Figure 3.



(a) $Z^{(e,1)}(\beta, \gamma)$ (b) $C^{(e,1)}(\beta, \gamma)$ (c) $L^{(e,1)}(\beta, \gamma)$

(d) $Z^{(b,1)}(\epsilon, \gamma)$ (e) $C^{(b,1)}(\epsilon, \gamma)$ (f) $L^{(b,1)}(\epsilon, \gamma)$

Figure 3: Comparison of $\mathcal{D}_1$ Elastic Net and Berhu optimization

In an objective function with an Elastic Net penalty, the marginal cost of a trade y is $\gamma[\beta +$

$(1-\beta)y]$. With a Berhu penalty, it becomes γ for $y \leq \epsilon$, and $\frac{\gamma y}{\epsilon}$ elsewhere. In Elastic Nets, the composite coefficient on the $L1$ penalty is $\gamma\beta$, which applies to the entire range of y. Meanwhile, an increase in γ in a Berhu penalty is reflected directly on the L_1 penalty as long as y falls below ϵ, beyond which the penalty is on the $L2$. This difference is reflected in the plots of $Z^{(e,1)}$ and $Z^{(b,1)}$ in Figure 3. In general, $Z^{(e,1)}$ is not very sensitive to the choice β except when β is below 0.5.

Where $Z^{(e,1)}$ is insensitive to γ, a bigger γ still negatively impacts $C^{(e,1)}$ and $L^{(e,1)}$, continuously pushing the solution away from the target. Therefore, a sensible strategy calls for a reasonably large β while keeping γ as small as possible in an Elastic Net penalty.

That is very different from the Berhu cases, where a large γ increases the sensitivity of $Z^{(b,1)}$ to ϵ, decreases the sensitivities of both $C^{(b,1)}$ and $L^{(b,1)}$. Overall, $Z^{(b,1)}$ is very sensitive to ϵ across all the choices of γ. However, a small ϵ may cause $Z^{(b,1)}$ to stay below 0, meaning that the penalty increases number of noise trades. A small ϵ will subject an increasing number of trades to $L2$ penalty, spreading small trades instead of pruning them. Therefore, it is critical to choose an ϵ that makes $L1$ penalty effective over a decent portion of a portfolio. With an appropriate ϵ, $C^{(b,1)}$ has a larger flat area of relatively high values, meaning less correlation difference from the baseline values. This implies that a proper choice of ϵ affords more room to find a superior solution to the Elastic Net as long as only a modest number of noise trades need to be pruned.

6 Concluding Remarks

The penalized portfolio optimization techniques studied in this paper address two deficiencies of the conventional approach. The first one concerns rigid constraints that sometimes end up driving the solution. The Alpha Gate introduced in Section 3 directly incorporates violation of a soft threshold as a penalty term in an objective function. It can be viewed as a flexible constraint that can be stretched to some degree. The second one serves as a shrinkage mechanism, similar to variable selection in regression settings. The regularization methods introduced in Section 4 aim to eliminate noisy trades.

A main concern with penalty terms is that they tend to add biases to original objective functions. This is particularly true in the case of regularization methods. One way to mitigate the bias is to follow a common practice in regression studies. After identifying relevant features, a regression is often run with only the selected features. In the portfolio optimization, the noise trades that fall below a tiny threshold from a penalized optimization can be forced to zero and another optimization is then performed without such a penalty term. This protects the optimal solution from bias.

Another way to mitigate penalty biases takes the form of non-convex penalty functions. The SCAD penalty by [4] and the MCP penalty by [13] can be applied to $|w-w^*|$ to penalize turnovers. Like Elastic Net and Berhu, both penalty functions are non-differentiable at zero, allowing for sparsity. But unlike those convex penalties, these are concave elsewhere. Consequently, the marginal penalty reaches zero as $|w-w^*|$ increases and the total penalty stops increasing. Effectively, such penalty forms set a ceiling on the penalty biases. However, non-convexity of these functions makes

it very difficult to optimize.

More systematic and objective procedures to select penalty parameters also warrant further research.

Another line of research studies alternative forms of loss functions. The Huber's Loss function (13) is particularly interesting as an alternative objective function. For an individual tracking error $x_i = w_i - \tilde{w}_i$, and a threshold parameter δ, the Huber's Loss function can be written as

$$h(x_i) = \begin{cases} \frac{x_i^2}{2\delta} & \text{if } |x_i| \leq \delta, \\ x_i - \frac{\delta}{2} & \text{if } |x_i| > \delta. \end{cases}$$

Such a loss function dampens the larger points beyond the threshold δ. It reduces impact of large deviations from the target portfolio and may indirectly lower the tendency of high turnovers.

Similar to (15), this can be reformulated to a single equation using auxiliary variables u_i and v_i, then scaled by a factor λ:

$$h(x_i) = \lambda \left(\frac{u_i^2}{2\delta} + v_i \right), \tag{18}$$

where

$$\begin{aligned} u_i + v_i &\geq |x_i|, \\ 0 \leq u_i &\leq \delta, \\ v_i &\geq 0. \end{aligned}$$

CVXPY [3] contains designated Huber functions that allow direct specification without slack vectors u and v. Behind the scenes, however, it carries out the similar conversions from Huber functions.

The loss function (18) can be made the objective function, along with other penalties such as turnover gating, Elastic Net, or Berhu Loss. In portfolio optimization, minimizing the squared tracking error is sensitive to outliers. The Huber's Loss based objective function penalizes larger $|w - \tilde{w}|$ less than the quadratic one. The value of λ dictates how much the solver discourages responses to large outliers in $\tilde{w}$ above δ.

Appendix A: Additional Graphs

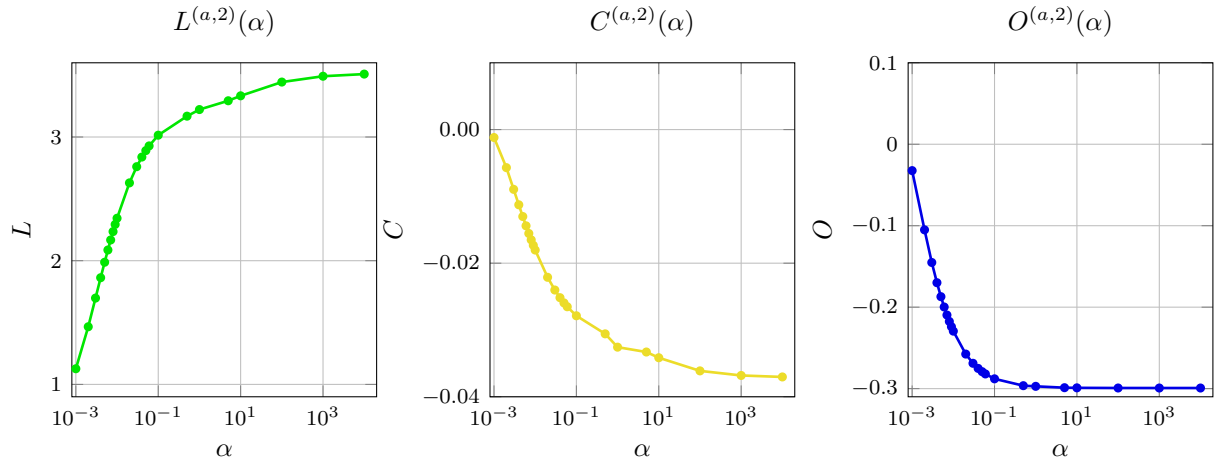


Figure 4: Alpha Gating $\mathcal{D}_2$

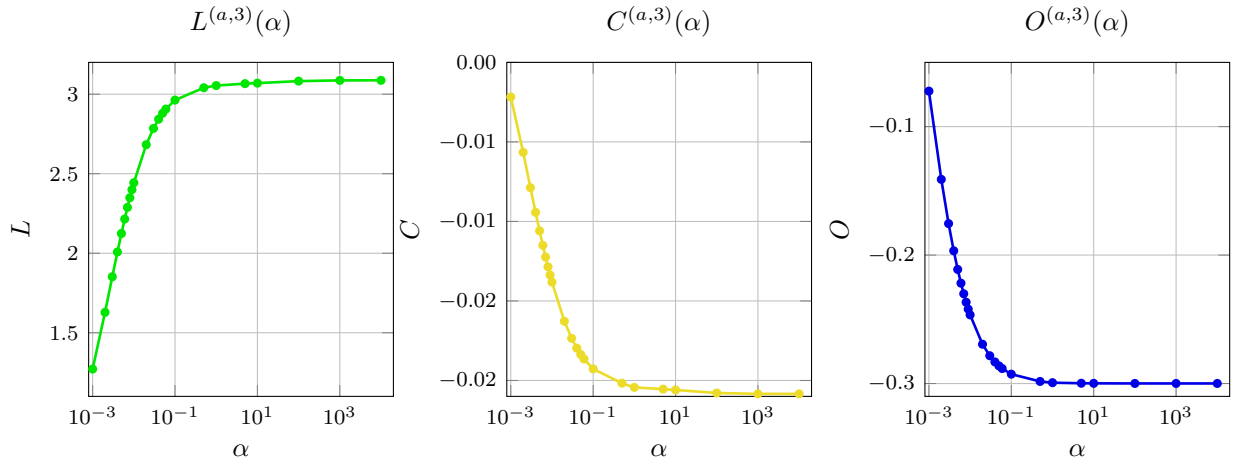
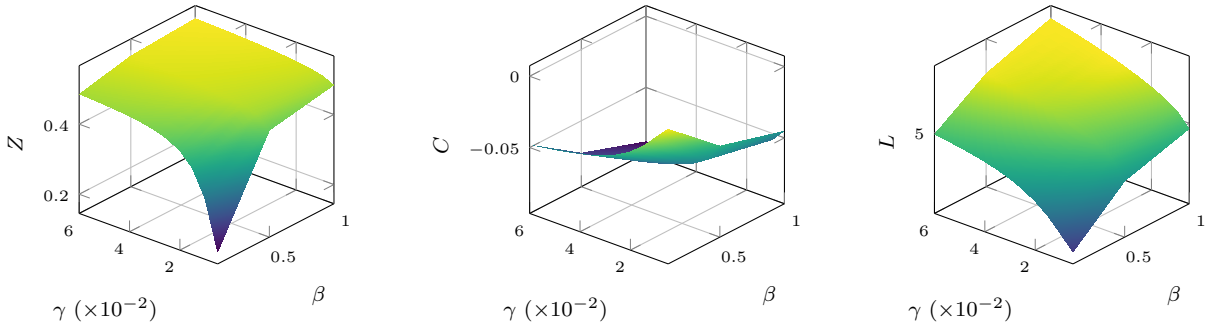
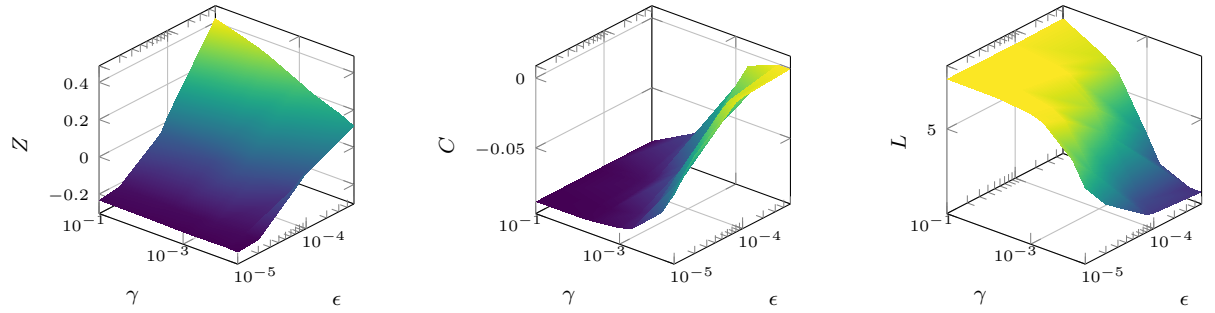


Figure 5: Alpha Gating $\mathcal{D}_3$

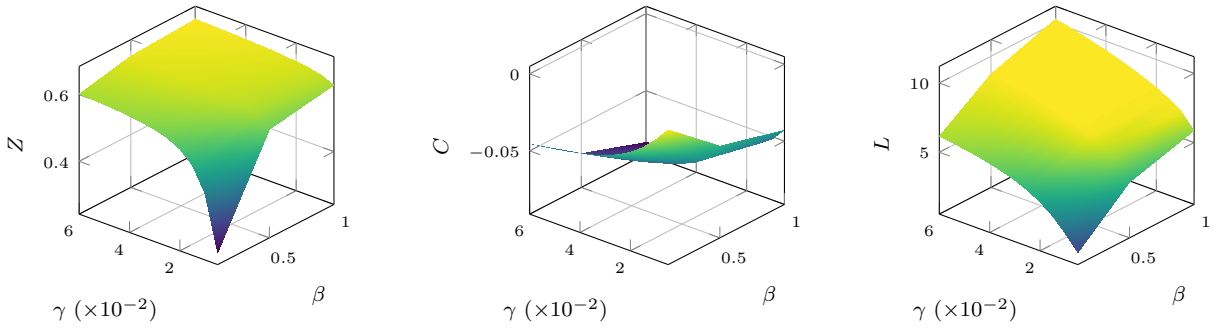


(a) $Z^{(e,2)}(\beta,\gamma)$ (b) $C^{(e,2)}(\beta,\gamma)$ (c) $L^{(e,2)}(\beta,\gamma)$

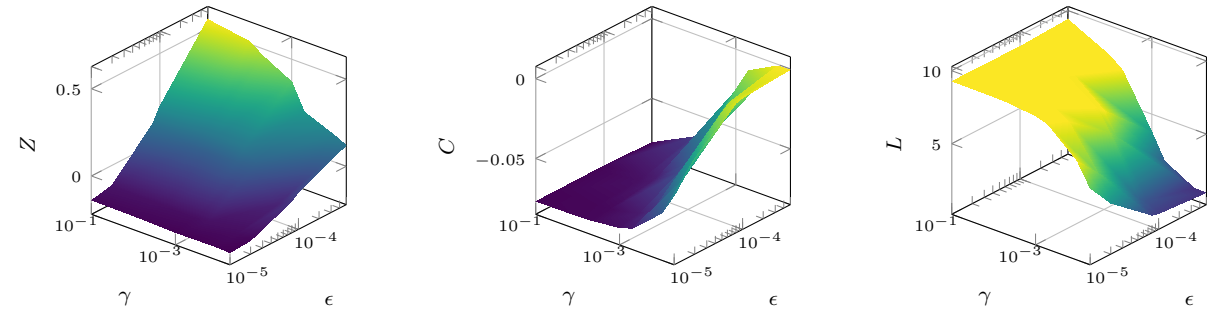


(d) $Z^{(b,2)}(\epsilon,\gamma)$ (e) $C^{(b,2)}(\epsilon,\gamma)$ (f) $L^{(b,2)}(\epsilon,\gamma)$

Figure 6: Comparison of $\mathcal{D}_2$ Elastic Net and Berhu optimization



(a) $Z^{(e,3)}(\beta,\gamma)$ (b) $C^{(e,3)}(\beta,\gamma)$ (c) $L^{(e,3)}(\beta,\gamma)$



(d) $Z^{(b,3)}(\epsilon,\gamma)$ (e) $C^{(b,3)}(\epsilon,\gamma)$ (f) $L^{(b,3)}(\epsilon,\gamma)$

Figure 7: Comparison of $\mathcal{D}_1$ Elastic Net and Berhu optimization

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